# STAFF REVIEW · H446 186 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

## Do Now — source answers

1. They shift one position right to make space.

2. [1,3,4].

3. 4.

4. Result: [2,4,5,6,8]. Shift 8, then 6 right; insert 5 after 4.

## I Do — source worked steps

1. Split until each call is trivial — Split [8,3,5,2] into [8,3] and [5,2], then into [8],[3],[5],[2]. Each length-one call returns unchanged; an empty input also returns immediately.

2. Return through the merges — Merge [8] with [3]→[3,8]; [5] with [2]→[2,5]. Merge fronts 2,3,5, then copy8: [2,3,5,8]. Each helper input is already sorted. The outline uses end-exclusive copyRange and the merge loop above.

## We Do — guided CHECK

1. Check the complete trace
Singletons return; merge [1,6] and [2,4]; final [1,2,4,6]. copyRange copies the stated slice, excluding stop; MergeSorted compares fronts and copies the remaining tail. These are defined helpers, not built-in OCR commands.

## You Do — source mark schemes

1. [1,2,5,6].

2. [4,1]|[3,2] → [4],[1],[3],[2]. Length 0/1 is the base case. Return merges [1,4] and [2,3], then [1,2,3,4].

3. There are about $\log_2 n$ merge levels, and each level processes n elements → O(n log n) time, with O(n) extra array storage for this implementation.

## Exit Ticket — source answer

1. False. Both input runs must already be sorted.